

Network and System Management

Level 3: Step 1 normal - Let's go net fishing

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Question 1: BIND9 Zone Files

BIND9 DNS Server stores DNS records in zone files that map hostnames to IP addresses (forward lookup) and IP addresses to hostnames (reverse lookup).

Forward Lookup Zone

The forward lookup zone maps hostnames to IP addresses. It contains SOA (Start of Authority) records, NS (Name Server) records, and A (Address) records. The SOA record includes the serial number used for zone transfer synchronization. Example syntax for linux.local domain:

```
$TTL      604800 @          IN          SOA      l3s1n-dns1.linux.local. admin.linux.local. (
2          ; Serial                          604800          ; Refresh
86400      ; Retry                            2419200         ; Expire
604800 )    ; Negative Cache TTL @          IN          NS      l3s1n-dns1.linux.local. @
IN          NS      l3s1n-dns2.linux.local. l3s1n-dns1      IN          A          192.168.8.200
l3s1n-dns2   IN          A          192.168.8.201 l3s1n-dhcp      IN          A
192.168.8.100 l3s1n-wstation IN          A          192.168.8.90
```

Reverse Lookup Zone

The reverse lookup zone maps IP addresses to hostnames using PTR (Pointer) records. The zone name is derived from the subnet in reverse order with .in-addr.arpa appended. Example for 192.168.8.0/24 subnet:

```
$TTL      604800 @          IN          SOA      l3s1n-dns1.linux.local. admin.linux.local. (
2          ; Serial                          604800          ; Refresh
86400      ; Retry                            2419200         ; Expire
604800 )    ; Negative Cache TTL @          IN          NS      l3s1n-dns1.linux.local.
200      IN          PTR      l3s1n-dns1.linux.local. 201      IN          PTR      l3s1n-
dns2.linux.local. 100      IN          PTR      l3s1n-dhcp.linux.local. 90      IN          PTR
l3s1n-wstation.linux.local.
```

Question 2: Zone Transfer Between BIND9 Servers

Zone transfer is the mechanism by which BIND9 secondary (slave) DNS servers obtain copies of zone data from primary (master) servers to provide redundancy and load distribution.

When Zone Transfer Occurs

Zone transfers happen automatically when: (1) the secondary server starts, (2) the refresh timer expires, (3) the primary sends a NOTIFY message after zone updates, or (4) the serial number on the primary exceeds the secondary's serial number.

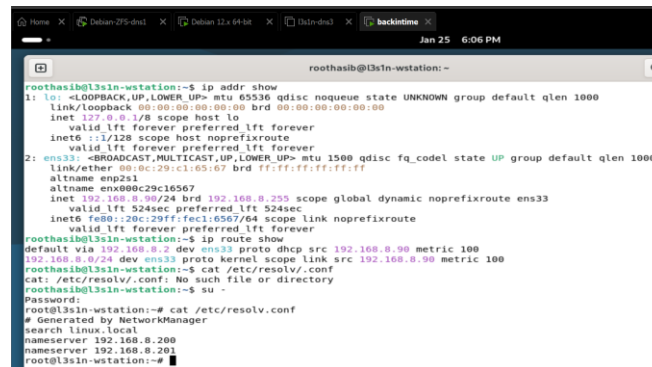
How Zone Transfer Works

The primary server is configured with allow-transfer directives specifying authorized secondary servers. The secondary server is configured with zone type 'slave' and the primary's IP in the masters directive. When zone data changes, the primary sends a NOTIFY message to secondaries. The secondary queries the primary's SOA record to check the serial number. If the primary's serial is higher, the secondary requests a zone transfer (AXFR or IXFR) over TCP port 53. The secondary receives the complete zone data and stores it locally, making it available for DNS queries.

Question 3: Practical Implementation

3(a): VM Creation and Network Information

Four virtual machines were created in VMware's NAT network: l3s1n-dns1 (Debian 11), l3s1n-dns2 (Debian 11), l3s1n-dhcp (Debian 11), and l3s1n-wstation (Ubuntu Desktop). Network information was retrieved showing subnet 192.168.8.0/24, gateway 192.168.8.2, and DNS server 192.168.8.2.

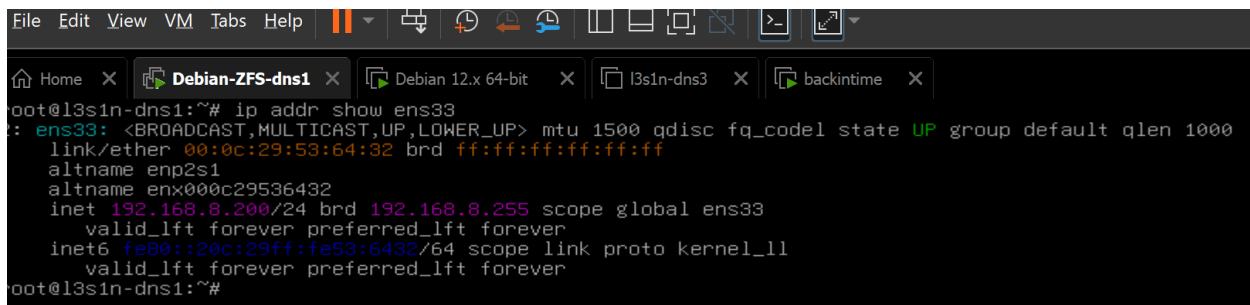


```
root@l3s1n-wstation:~# ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c1:65:67 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29c16567
    inet 192.168.8.96/24 brd 192.168.8.255 scope global dynamic noprefixroute ens33
        valid_lft 524sec preferred_lft 524sec
    inet6 fe80::20c:29ff:fe1:6567/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@l3s1n-wstation:~# ip route show
default via 192.168.8.2 dev ens33 proto dhcp src 192.168.8.96 metric 100
192.168.8.0/24 dev ens33 proto kernel scope link src 192.168.8.96 metric 100
root@l3s1n-wstation:~# cat /etc/resolv.conf
cat: /etc/resolv.conf: No such file or directory
root@l3s1n-wstation:~# su -
Password:
root@l3s1n-wstation:~# cat /etc/resolv.conf
# Generated by NetworkManager
search linux.local
nameserver 192.168.8.200
nameserver 192.168.8.201
root@l3s1n-wstation:~#
```

Figure 1: Network information retrieved using `ip addr show` and `ip route show`

3(b): Static IP Configuration

Static IP addresses were configured on the three Debian server VMs by editing `/etc/network/interfaces`. The configuration set l3s1n-dns1 to 192.168.8.200, l3s1n-dns2 to 192.168.8.201, and l3s1n-dhcp to 192.168.8.100, with appropriate netmask, gateway, and DNS server settings.



```
File Edit View VM Tabs Help
root@l3s1n-dns1:~# ip addr show ens33
ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:53:64:32 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c29536432
    inet 192.168.8.200/24 brd 192.168.8.255 scope global ens33
        valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe53:6432/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
root@l3s1n-dns1:~#
```

Figure 2: Static IP addresses configured on all three servers

3(c): Disabling VMware Internal DHCP

The VMware Workstation internal DHCP server was disabled for the NAT network (VMnet8) through the Virtual Network Editor. This allows the custom ISC DHCP server to manage IP address assignment without conflicts.

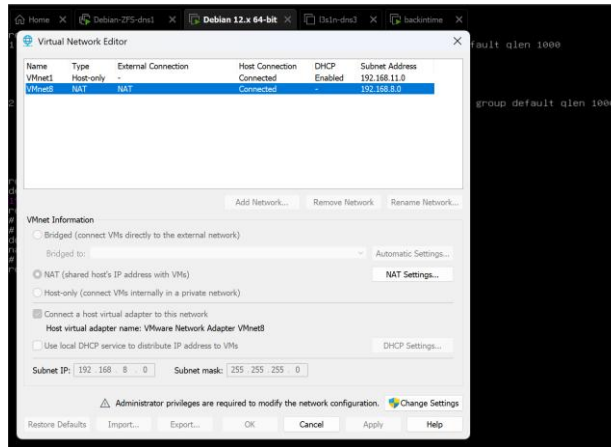


Figure 3: VMware Virtual Network Editor showing DHCP disabled on VMnet8

3(d): BIND9 DNS Server Configuration

```

root@l3s1n-dns1:~# systemctl status bind9
* named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: inactive (dead) since Sun 2026-01-25 21:55:13 HET; 2h 29min ago
   Duration: 1h 43min 28.782s
   Invocation: b78f9438bf9492faadad734aa355c65
   Docs: man:named(8)
   Process: 2243 ExecStart=/usr/sbin/named -f $OPTIONS (code=killed, signal=TERM)
   Process: 2281 ExecStop=/usr/sbin/rndc stop (code=exited, status=0/SUCCESS)
   Main PID: 2243 (code=killed, signal=TERM)
   Status: "running"
   Mem peak: 40.7M
   CPU: 556ms

Jan 25 21:55:13 l3s1n-dns1 named[2243]: received control channel command 'stop'
Jan 25 21:55:13 l3s1n-dns1 named[2243]: no longer listening on 127.0.0.1#53
Jan 25 21:55:13 l3s1n-dns1 named[2243]: no longer listening on 192.168.0.200#53
Jan 25 21:55:13 l3s1n-dns1 named[2243]: no longer listening on ::1#53
Jan 25 21:55:13 l3s1n-dns1 named[2243]: no longer listening on fe80::2ac:29ff:fe53:6432#53
Jan 25 21:55:13 l3s1n-dns1 named[2243]: stopping command channel on 127.0.0.1#953
Jan 25 21:55:13 l3s1n-dns1 named[2243]: stopping command channel on ::1#953
Jan 25 21:55:13 l3s1n-dns1 named[2243]: shutting down: flushing changes
Jan 25 21:55:13 l3s1n-dns1 systemd[1]: named.service: Deactivated successfully.
Jan 25 21:55:13 l3s1n-dns1 systemd[1]: Stopped named.service - BIND Domain Name Server.
root@l3s1n-dns1:~#

root@l3s1n-dns1:~# cat /etc/bind/db.linux.local
$
IN      SOA      l3s1n-dns1.linux.local. admin.linux.local. (
        604800      : Serial
        604800      : Refresh
        86400       : Retry
        2419200     : Expire
        604800 )    : Negative Cache TTL
;
$
IN      NS       l3s1n-dns1.linux.local.
$
IN      NS       l3s1n-dns2.linux.local.
l3s1n-dns1 IN  A      192.168.0.200
l3s1n-dns2 IN  A      192.168.0.201
l3s1n-dhcp IN  A      192.168.0.100
l3s1n-wstation IN A      192.168.8.90
root@l3s1n-dns1:~# cat /etc/bind/db.192.168.8
$
IN      SOA      l3s1n-dns1.linux.local. admin.linux.local. (
        604800      : Serial
        604800      : Refresh
        86400       : Retry
        2419200     : Expire
        604800 )    : Negative Cache TTL
;
$
IN      NS       l3s1n-dns1.linux.local.
$
IN      NS       l3s1n-dns2.linux.local.
$
IN      PTR      l3s1n-dns1.linux.local.
$
IN      PTR      l3s1n-dns2.linux.local.
$
IN      PTR      l3s1n-dhcp.linux.local.
$
IN      PTR      l3s1n-wstation.linux.local.

```

3(e): ISC DHCP Server Configuration

ISC DHCP Server was installed on l3s1n-dhcp and configured to serve the 192.168.8.0/24 subnet. The configuration provides dynamic IP assignment in the range 192.168.8.10-192.168.8.50, sets the domain name to linux.local, configures DNS servers as 192.168.8.200 and 192.168.8.201, and sets the gateway to 192.168.8.2. A host reservation was created for l3s1n-wstation (MAC address 00:0c:29:c1:65:67) to receive fixed IP 192.168.8.90.

```

root@l3sin-dhcp:~# systemctl status isc-dhcp-server
* isc-dhcp-server.service - LSB: DHCP server
   Loaded: loaded (/etc/init.d/isc-dhcp-server; generated)
   Active: active (running) since Mon 2025-01-26 01:41:40 HET; 6min ago
   Invocation: af3d312859d240c3b9ac758c5256ee9a
   Docs: man:systemd-sysv-generator(8)
   Process: 5230 ExecStart=/etc/init.d/isc-dhcp-server start (code=exited, status=0/SUCCESS)
   Tasks: 1 (limit: 2267)
   Memory: 7.3M (peak: 9.6M)
   CPU: 261ms
   CGroup: /system.slice/isc-dhcp-server.service
           └─5242 /usr/sbin/dhcdp -4 -q -cf /etc/dhcp/dhcpd.conf ens33

Jan 26 01:41:37 l3sin-dhcp dhcdp[5242]: Wrote 0 deleted host decls to leases file.
Jan 26 01:41:37 l3sin-dhcp dhcdp[5242]: Wrote 0 new dynamic host decls to leases file.
Jan 26 01:41:37 l3sin-dhcp dhcdp[5242]: Wrote 0 leases to leases file.
Jan 26 01:41:38 l3sin-dhcp dhcdp[5242]: Server starting service.
Jan 26 01:41:40 l3sin-dhcp isc-dhcp-server[5230]: Starting ISC DHCPv4 server: dhcdp.
Jan 26 01:41:40 l3sin-dhcp systemd[1]: Started isc-dhcp-server.service - LSB: DHCP server.
Jan 26 01:41:45 l3sin-dhcp dhcdp[5242]: DHCPREQUEST for 192.168.8.90 from 00:0c:29:c1:65:67 via ens33
Jan 26 01:41:45 l3sin-dhcp dhcdp[5242]: DHCPACK on 192.168.8.90 to 00:0c:29:c1:65:67 via ens33
Jan 26 01:46:45 l3sin-dhcp dhcdp[5242]: DHCPREQUEST for 192.168.8.90 from 00:0c:29:c1:65:67 via ens33
Jan 26 01:46:45 l3sin-dhcp dhcdp[5242]: DHCPACK on 192.168.8.90 to 00:0c:29:c1:65:67 via ens33
root@l3sin-dhcp:~# echo "DHCP Configuration"
DHCP Configuration
root@l3sin-dhcp:~# cat /etc/dhcp/dhcpd.conf | tail -15
subnet 192.168.8.0 netmask 255.255.255.0 {
    range 192.168.8.10 192.168.8.50;
    option domain-name "linux.local";
    option domain-name-servers 192.168.8.200, 192.168.8.201;
    option routers 192.168.8.2;
    default-lease-time 600;
    max-lease-time 7200;
}

host l3sin-wstation {
    hardware ethernet 00:0c:29:c1:65:67;
    fixed-address 192.168.8.90;
}

root@l3sin-dhcp:~#

```

Figure 5: ISC DHCP Server running and configuration file contents

3(f): Workstation DHCP Configuration

The workstation (l3s1n-wstation) was configured to obtain IP and DNS addresses automatically via DHCP. The workstation successfully received the fixed IP address 192.168.8.90, domain name linux.local, DNS servers 192.168.8.200 and 192.168.8.201, and gateway 192.168.8.2 as configured on the DHCP server.

```

root@l3sin-wstation:~# hostname
l3sin-wstation
root@l3sin-wstation:~# ip addr show ens33
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c1:65:67 brd ff:ff:ff:ff:ff:ff
        altname enp2s1
        altnam enx000c29c16567
        inet 192.168.8.90/24 brd 192.168.8.255 scope global dynamic noprefixroute ens33
            valid lft 375sec preferred lft 375sec
        inet6 fe80::20c:29ff:fe01:6567/64 scope link noprefixroute
            valid lft forever preferred lft forever
root@l3sin-wstation:~# cat /etc/resolv.conf
# Generated by NetworkManager
search linux.local
nameserver 192.168.8.200
nameserver 192.168.8.201
root@l3sin-wstation:~# ip route show
default via 192.168.8.2 dev ens33 proto dhcp src 192.168.8.90 metric 100
192.168.8.0/24 dev ens33 proto kernel scope link src 192.168.8.90 metric 100
root@l3sin-wstation:~#

```

Figure 6: Workstation showing IP 192.168.8.90, domain linux.local, and DNS servers

3(g): DNS Resolution Testing

DNS resolution was tested using nslookup from the workstation. All queries were successful: (1) l3s1n-dhcp resolved to 192.168.8.100, (2) l3s1n-wstation resolved to 192.168.8.90, (3) reverse lookup of 192.168.8.100 returned l3s1n-dhcp.linux.local, and (4) www.google.com resolved to external IP addresses. All queries were answered by the primary DNS server at 192.168.8.200.



```
root@l3s1n-wstation:~# nslookup l3s1n-dhcp
Server:      192.168.8.200
Address:     192.168.8.200#53

Name:   l3s1n-dhcp.linux.local
Address: 192.168.8.100

root@l3s1n-wstation:~# nslookup l3s1n-wstation
Server:      192.168.8.200
Address:     192.168.8.200#53

Name:   l3s1n-wstation.linux.local
Address: 192.168.8.90

root@l3s1n-wstation:~# nslookup 192.168.8.100
100.8.168.192.in-addr.arpa    name = l3s1n-dhcp.linux.local.

root@l3s1n-wstation:~# nslookup www.google.com
;; Got SERVFAIL reply from 192.168.8.200, trying next server
Server:      192.168.8.201
Address:     192.168.8.201#53

Non-authoritative answer:
Name:   www.google.com
Address: 172.217.20.228
Name:   www.google.com
Address: 2a00:1450:4003:809::2004

root@l3s1n-wstation:~#
```

Figure 7: DNS resolution testing showing successful forward, reverse, and external lookups

3(h): Secondary DNS Failover Testing

DNS redundancy was verified by stopping the BIND9 service on the primary server (l3s1n-dns1). The same DNS queries were executed from the workstation. The queries showed connection refused errors to 192.168.8.200 (primary server down), but all queries were successfully answered by the secondary DNS server at 192.168.8.201. This confirms proper zone transfer occurred and the secondary server can handle DNS queries when the primary is unavailable.



```
root@l3s1n-wstation:~# nslookup l3s1n-dhcp
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused
Server:      192.168.8.201
Address:     192.168.8.201#53

Name:   l3s1n-dhcp.linux.local
Address: 192.168.8.100
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused

root@l3s1n-wstation:~# nslookup www.google.com
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused
Server:      192.168.8.201
Address:     192.168.8.201#53

Non-authoritative answer:
Name:   www.google.com
Address: 172.217.20.228
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused
;; communications error to 192.168.8.200#53: connection refused
Name:   www.google.com
Address: 2a00:1450:4003:809::2004

root@l3s1n-wstation:~#
```

Figure 8: DNS queries successful with primary server down, answered by secondary server

Conclusion

The implementation successfully demonstrates a complete DNS and DHCP infrastructure with proper redundancy. The primary-secondary DNS configuration ensures high availability through zone transfer, while the DHCP server provides automated network configuration with both dynamic and static IP assignment capabilities. All services were verified to be functioning correctly through comprehensive testing, including failover scenarios.